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Understanding and addressing quality attributes of microservices architecture: A Systematic literature review

Shanshan Li [a], He Zhang [a,*], Zijia Jia [a], Chenxing Zhong [a], Cheng Zhang [b], Zhihao Shan [c], Jinfeng Shen [a], Muhammad Ali Babar [d]

[a] State Key Laboratory of Novel Software Technology Software Institute, Nanjing University, Jiangsu, China
[b] School of Computer Science & Technology, Anhui University, Anhui, China
[c] Tencent Technology Co.Ltd, Shenzhen, China
[d] CREST, University of Adelaide, Adelaide, Australia

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ABSTRACT

Context: As a rapidly adopted architectural style in software engineering, Microservices Architecture (MSA) advocates implementing small-scale and independently distributed services, rather than binding all functions into one monolith. Although many initiatives have contributed to the quality improvement of microservices-based systems, there is still a lack of a systematic understanding of the Quality Attributes (QAs) associated with MSA.

Objective: This study aims to investigate the evidence-based state-of-the-art of QAs of microservices-based systems.

Method: We carried out a Systematic Literature Review (SLR) to identify and synthesize the relevant studies that report evidence related to QAs of MSA.

Results: Based on the data extracted from the 72 selected primary studies, we portray an overview of the six identified QAs most concerned in MSA, *scalability, performance, availability, monitorability, security*, and *testability*. We identify 19 tactics that architecturally address the critical QAs in MSA, including two tactics for *scalability*, four for *performance*, four for *availability*, four for *monitorability*, three for *security*, and two for *testability*.

Conclusion: This SLR concludes that for MSA-based systems: 1) Although *scalability* is the commonly acknowledged benefit of MSA, it is still an indispensable concern among the identified QAs, especially when trading-off with other QAs, e.g., *performance*. Apart from the six identified QAs in this study, other QAs for MSA like *maintainability* need more attention for effective improvement and evaluation in the future. 3) Practitioners need to carefully make the decision of migrating to MSA based on the return on investment, since this architectural style additionally cause some pains in practice.

1. Introduction

Microservices Architecture (MSA) originates from agile developer communities and has received shifting focus in both industry and academia in recent years [1] after the prosperity of SOA (Service--oriented Architecture). No agreement has been reached about the relationship between MSA and SOA [2]. Some MSA advocates claim that MSA is a new architectural style, while some SOA proponents think that MSA is merely an implementation approach of SOA. Our paper follows the former claim based on the definition and nine characteristics of MSA from Fowler et al. [1]. and the seven principles defined by Newman [3].

MSA advocates decomposing the application into a set of small services and making them communicate with each other through light weight mechanisms (e.g., RESTful API or stream-based communications) [1]. Typical characteristics and benefits [2] of MSA are "componentization via services" improving *modifiability, scalability*, and *deployability*, "organized around business capabilities" making code understandable and easy to maintain, "infrastructure automation" for continuous delivery and DevOps, and "Decentralized governance and data management" enabling polyglot persistence promotes flexibility and suitability, etc. Inspired by the perceived benefits of MSA, many world-leading Internet enterprises such as Netflix, Amazon and eBay

* Corresponding author.
E-mail address: dr.hezhang@gmail.com (H. Zhang).

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